

PROFESSIONAL SUMMARY

Motivated Software Tools Engineer with an Integrated M.Tech in Computer Science (Cybersecurity) and a strong foundation in **Python development**, **Linux operating systems**, and **AI/ML model integration**. Proven ability to build automated systems for anomaly detection and system monitoring—directly transferable to QA automation and test case generation. Passionate about system architecture, PC hardware interfaces, and developing scalable tools to optimize complex workflows. Eager to apply AI-driven methodologies to automate validation of device drivers and SDKs in a fast-paced, hardware-software co-design environment.

- EDUCATION
- VIT Bhopal University

Integrated M.Tech in Computer Science (Cybersecurity); CGPA: 8.01

Relevant Courses: Operating Systems, Computer Architecture, Linux System Programming, Data Structures & Algorithms, Database Management Systems, Network Security

India

2023 – 2028
- School of Scholars

Higher Secondary Education (11th & 12th); 85%, 76%
- India
- 2021 – 2023

- SKILLS SUMMARY
- Languages:

Python (Expert – OOP, Dictionaries, Tuples, RegEx, PIP), C, C++, SQL, Bash/Shell Scripting, Java, GoLang

AI & Test Automation:

ML-based Anomaly Detection, Predictive Failure Analysis, AI-driven Log Analysis, Automated Alerting, Scikit-learn, Pandas, NumPy

Frameworks & APIs:

PyTest, Robot Framework, Flask, Django, REST APIs

OS & System Architecture:

Linux (Kernel Fundamentals, Process/Thread Management, File Systems, Performance Tuning), Unix, PC Hardware Architecture, PCI-E, USB, SATA, Device Driver Validation Concepts, GDB, Valgrind

Databases & Storage:

MongoDB, Elasticsearch, MySQL, PostgreSQL

DevOps & CI/CD Tools:

Git, Jenkins, GitLab CI, Docker, JIRA, Confluence, Code Coverage (gcov/lcov), Static/Dynamic Analysis Tools

- EXPERIENCE
- COEP Technological University

Research Intern

Data Engineering & Visualization: Built Python pipelines (Pandas, NumPy) for geospatial datasets and interactive dashboards, delivering actionable insights for environmental analysis.

Systematic Validation: Applied rigorous analytical methods to evaluate stress factors, mirroring root-cause analysis and validation workflows for GPU testing.

Pune, India (On-Site)

May 2025 – Jul 2025
- MPPolice

Cybersecurity Intern

Log Analysis & Automation: Analyzed Linux-based system logs (RegEx), developed Bash/Python scripts for security audits, and automated alert generation—directly applicable to test log analysis and defect detection.

System Hardening: Enhanced digital security frameworks and OS configurations, demonstrating deep Linux fundamentals and system tuning expertise.
- India (Hybrid)
- Sep 2024 – Oct 2025

- PROJECTS
- AI-Driven Test Anomaly Detection Framework:

Built an ML classification system (Random Forest, Logistic Regression) in Python to detect system anomalies in real-time, simulating device driver validation. Integrated Elasticsearch for centralized logging and automated alerts, showcasing end-to-end QA workflow design and AI-driven automation.

Distributed System & Monitoring Tool:

Developed a secure client-server architecture (Python, OOP, AES/HMAC encryption) with an endpoint agent to track file/process/network activity on Linux. Implemented automated responses and a central dashboard, mirroring robust tool development for test frameworks.

Predictive Analytics & Reporting System:

Created a time-series ML pipeline (Pandas, Scikit-learn) to forecast system loads and CI/CD bottlenecks. Built a Flask-based dashboard for actionable reporting, demonstrating data preprocessing and stakeholder communication skills.

- HONORS AND AWARDS
- SIH India 2023 – College Finalist

NASSCOM Agentic AI Hackathon (Global) – Finalist

SolVIT Hackathon 2025 – 3rd Position amongst 150+ teams

- VOLUNTEER EXPERIENCE
- Android Club, VIT Bhopal

R&D Co-Lead

Team Leadership: Led a team of 10+ developers in research-driven projects focused on security and system-level tools.

Agile Workflows: Managed sprint planning, code reviews, and project deliverables, fostering a collaborative environment that mirrors NVIDIA’s peer-review and team-based culture.

Technical Mentorship: Conducted workshops on Python scripting, Linux fundamentals, and secure coding, refining my ability to estimate tasks and drive projects to closure.

India

Apr 2024 – Jul 2025